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SARDAR PATEL UNIVERSITY
External Examination
M. Sc. – Information Technology (Integrated) – Sixth Semester
PS06CIIT03: Computer Graphics
7th April, Thursday, 2016

Time: 02:30pm to 05:30pm

Total Marks: 70

Q-1 Select an appropriate option.

10

- 1 _____ technique is used to modify or interpret existing pictures.
(a) Computer Art (b) Image Processing
(c) Visualization (d) None of the Above
- 2 _____ persistence phosphor is used in animation.
(a) Low (b) High (c) Medium (d) None of the Above.
- 3 _____ color display method is used in Random scan display.
(a) Shadow mask (b) Beam penetration
(c) Color display (d) None of the above.
- 4 DDA Stands for _____.
(a) Data Drawing Algorithm (b) Digital Drawing Algorithm
(c) Digital Differential Analyzer (d) Digital Drawing Analyzer
- 5 Odd-even rule also known as _____.
(a) Odd-parity rule (b) Even-parity rule
(c) Only (b) (d) Both (a) and (b)
- 6 In Non-zero winding rule, point is considered exterior if winding number is _____.
(a) 0 (b) Equal to edges of region (c) 1 (d) -1
- 7 Rotation point is also known as _____.
(a) Rotation angle (b) Rotation axis
(c) Pivot point (d) Translation vector
- 8 A _____ is a transformation that produces a mirror image of an object.
(a) Translation (b) Reflection (c) Rotation (d) Shear
- 9 Default background color of stage in FLASH is _____.
(a) red (b) blue (c) black (d) white
- 10 Shortcut key to convert to symbol in FLASH is _____.
(a) F6 (b) F8 (c) Shift F8 (d) Shift F6

Q-2 Answer the following questions. (Any TEN) 20

- 1 Define: (i) Aspect Ratio (ii) Resolution
- 2 State disadvantage of Beam Penetration method.
- 3 Explain in-line arrangement electron gun in shadow mask method.
- 4 List various character attributes.
- 5 Differentiate: Bitmap font and Outline font.
- 6 Write only algorithm for Flood Fill operation.
- 7 What is scaling?
- 8 Give rotation matrix for 2-dimensional rotation transformation.
- 9 State condition for clipping a point.
- 10 Define: (i) Masking (ii) Layer
- 11 Explain timeline window.
- 12 State use of Eyedropper and Pan tool.

Q-3

- (a) Explain working of CRT in detail. 5
- (b) Differentiate between Raster scan displays and Random scan displays. 5

OR

Q-3

- (a) Write note on DVST. 5
- (b) List all types of input devices and explain any two in detail. 5

Q-4 Explain concept of line drawing and Write steps for Bresenham line drawing algorithm. 10

OR

Q-4 What is inside-outside test? Explain various methods to perform the same. 10

Q-5

- (a) What is geometric transformation? List all 2-D geometric transformation and explain any one in detail. 5
- (b) Explain window-to-viewport coordinate transformation. 5

OR

Q-5

- (a) What is text clipping? List out all types and explain it with example. 5
- (b) Write a short note on Viewing Pipeline. 5

Q-6

- (a) What is onion skinning? Explain the use of onion skinning in detail. 5
- (b) Explain frame-by-frame animation with example. 5

OR

Q-6

- (a) Explain how to import and export images using FLASH. 5
- (b) Explain motion tween animation with example. 5

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